
Abstract

Mental health has become a critical global issue, especially with the increasing prevalence of anxiety, depression, and other psychological disorders. Despite the growing need for mental health support, stigma and privacy concerns prevent many individuals from seeking professional help (Corrigan et al., 2014). Digital mental health platforms have emerged as a potential solution to address these barriers by offering anonymity and accessibility (Naslund et al., 2017). This study explores the development and impact of a secure, anonymous mental health care system designed for users in Sri Lanka. The system incorporates AI-driven features, peer support mechanisms, and verified professional interventions to ensure effective and confidential mental health assistance. Additionally, it leverages social platforms to encourage engagement while maintaining user privacy. Previous studies highlight the effectiveness of online mental health interventions in improving psychological well-being and reducing stigma (Andersson et al., 2019). This research contributes to the growing body of evidence supporting digital mental health solutions as an essential component of modern healthcare systems.

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Chapter 01

Introduction

Problem

Mental health concerns in Sri Lanka are escalating, with rising depression and suicide rates. Many individuals hesitate to seek help due to **cultural stigma, fear of judgment, lack of awareness, privacy concerns, and financial barriers**. Traditional support methods, such as psychiatrists and counselors, are often avoided, while government helplines like **1926 , 1333** face limitations. Younger individuals, in particular, are reluctant to use phone-based support due to privacy concerns and discomfort with direct communication. Additionally, **private hospitals and online consultations are costly**, making mental health services inaccessible to many.

Solution

A **secure, anonymous, and technology-driven online platform** will be developed to improve mental health accessibility in Sri Lanka. The platform will provide **professional guidance, peer support, and privacy-focused communication** to encourage help-seeking without stigma. It will be **free to access and sustained through donations**, ensuring affordability. **Gamification, AI-driven counseling, and awareness campaigns** will enhance engagement, while crisis management tools will strengthen intervention efforts. By **personalizing support for different user needs** and leveraging technology, this solution aims to **bridge gaps in traditional mental health services and promote a more inclusive support system**. It Solves all problems currently have in sri lanka related mental health.

Purpose

The solution aims to provide **secure, anonymous, and accessible mental health support**, overcoming **stigma, privacy concerns, lack of awareness and financial barriers** in Sri Lanka. By leveraging **technology and community engagement**, it seeks to **educate, support, and empower individuals**, ensuring **inclusive and sustainable mental health care for all**.

Users

- Mental Health professionals (Doctors, Psychologists, Councilor , Psychiatrists etc)
- Sri Lankan People (Proactive Seekers , Aware but Hesitant, Unaware Individuals ,Avoidant or Resistant ,Crisis Cases , Support Seekers , Caregivers & Allies)
- Admin

Chapter 02

System Analysis and Requirement Engineering

I Gathered Data Using These Strategies,

- Meet a Few Mental Health Professionals (Councilor, psychologist, Psychiatrists)
- Meet Close Friends Who Suffered from Mental Health Issues
- Meet Some People Who Overcome and Suffer, crisis cases Individuals
- Engage with Society
- I do some questionnaires through online , but people hesitate to contribute and it was unsuccessful.

After analysing all decided the requirements, figure 1.1

Functional Requirement	
Anonymous Login	Users can log in anonymously for privacy protection.
Mental Health Awareness Forum	Articles , resources related mental health and share option .
AI-Based Personal Suggestions	AI-driven recommendations to assess and improve mental health.
Engage with Professionals	Secure chat system to connect users with mental health experts.
Professional Registration	Verified professionals can sign up to provide support.
Doctor/User Dashboards	Separate interfaces for doctors and users to manage sessions.
Video Call for Prescription Verification	Secure video call feature for doctors to verify patients before issuing prescriptions.
Donation Model	A funding system to sustain the platform through donations.
Admin Panel	A centralized dashboard to manage users, professionals, forums, and donations.
Non-Functional Requirements	
Advanced Encryption	Secure data protection to ensure user confidentiality.
Security	Robust security measures to prevent unauthorized access.

Performance	Optimized platform to handle high traffic smoothly.
Usability	User-friendly design for easy navigation and accessibility.

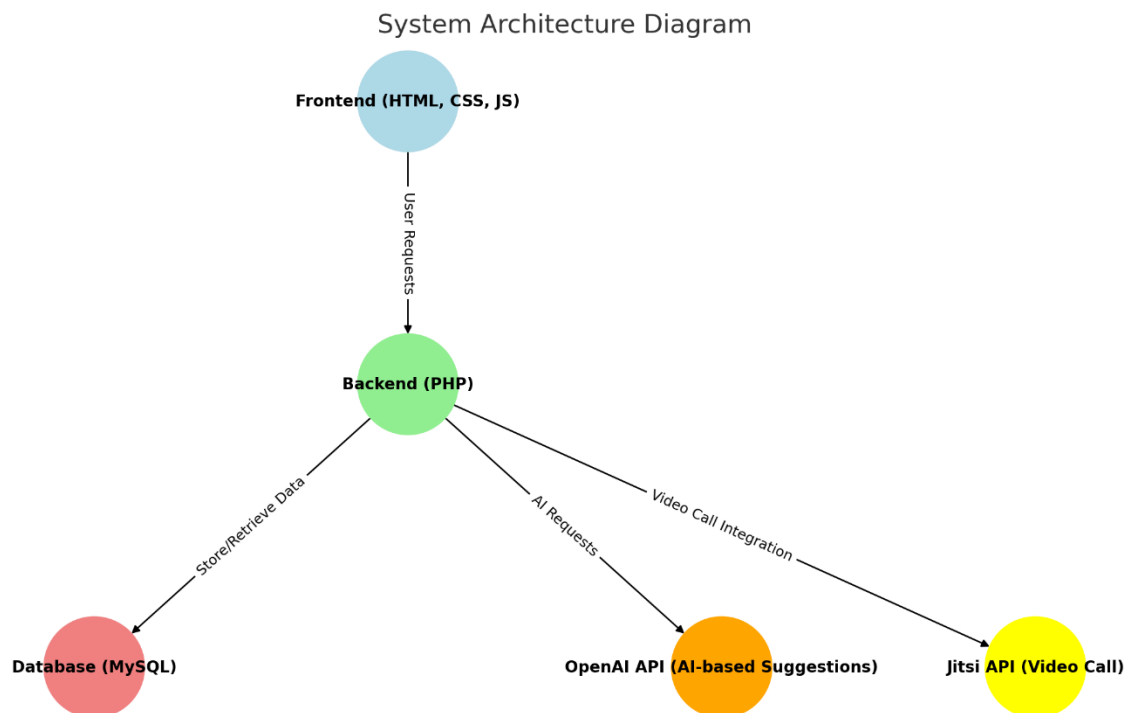
Chapter 03

System Architecture

Three-Tier Architecture - This structure enhances **performance, security, and scalability**, making the system **efficient and maintainable**.

- Frontend (Client-Side) – User interface using HTML, CSS, JavaScript.
- Backend (Server-Side) – Business logic implemented in PHP.
- Database (Storage) – SQL database for storing user data, forum discussions, and professional interactions.

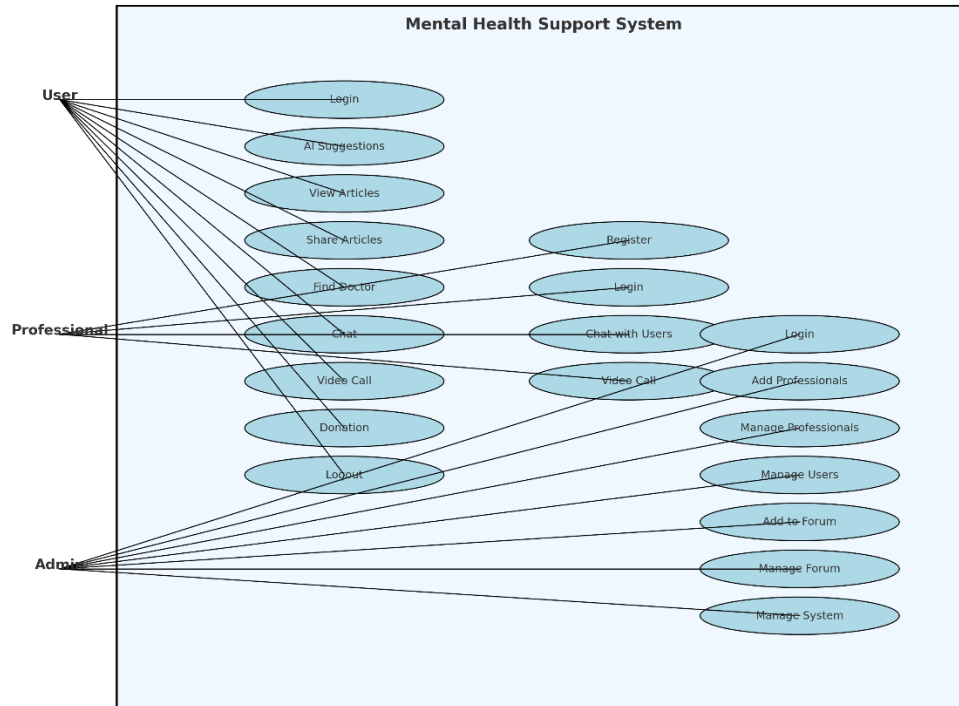
Used third-party services for accurate, fast features.



3.1

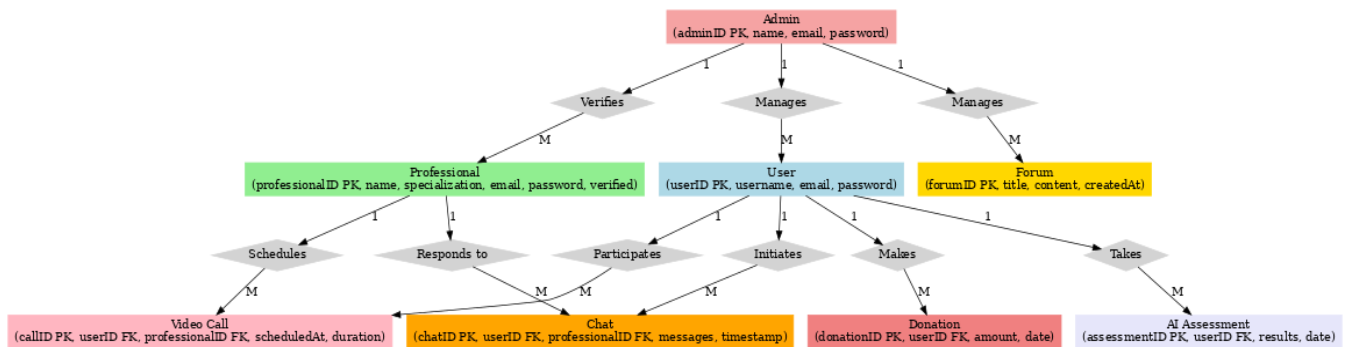
Use case diagram

Use Case Diagram - Mental Health Support System



3.2

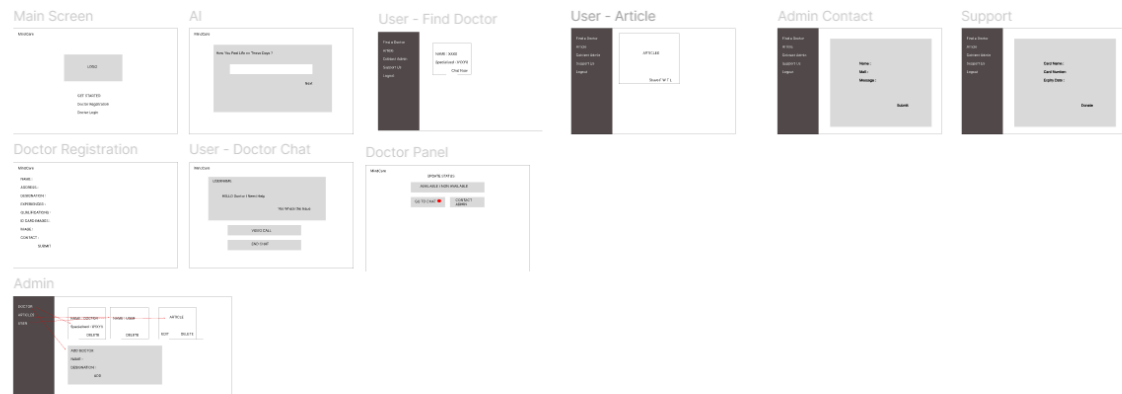
ER Diagram



3.3

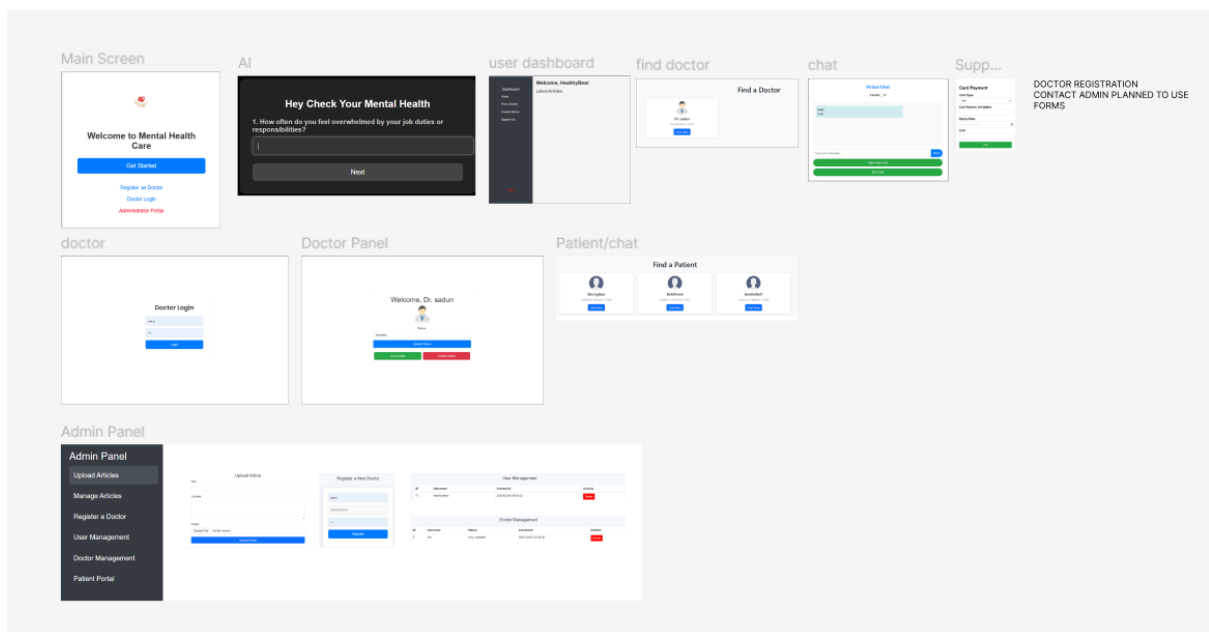
Wireframes :

<https://www.figma.com/design/nquz7GGGWjp4KQHqZ65Kve/Untitled?node-id=0-1&t=ur5t2wwlPBLqboTz-1>



3.4

UI : <https://www.figma.com/design/kCT62upRi53rglGH79SStu/Untitled?node-id=0-1&t=ZwCE3pKdQQjDhP43-1>



3.5

Chapter 04

Methodology

I Used Agile Methodology ,

Using **Agile** for your mental health support platform ensures **flexibility, continuous improvement, and user-centered development**. Agile enables **iterative development, quick adaptability to changes, and continuous feedback & testing**, ensuring **higher quality and faster delivery**. The methodology fosters **collaboration**, enhances **scalability**, and allows for **early release of core features** like chat, AI-based suggestions, and doctor consultations. This approach ensures the system remains **efficient, adaptable, and user-driven**, leading to a **successful and sustainable platform**.

Technology

- I Used Html , Css , Javascript for FrontEnd Development
- I Used MySql for Database
- I Used PHP for Backend
- As Third Party Services Use Jitsi for Video Calls , OpenAI for AI Feature

I used these technologies those are very easy to use , manage and provide advance feature with best optimization . Jitsi is free and very powerful. OpenAI is the world best AI it can analyse and provide better results .

Data Communication

I Used WIFI and API Calls for RealTime Updates and Functions (Video Call , AI Feature)

Using **WiFi and API calls** enables **real-time updates for video calls and AI features**, ensuring **fast, reliable communication, seamless doctor-patient interactions, quick AI responses, and system scalability**. This approach enhances **user experience, efficiency, and accessibility** of the mental health support platform.

Softwares

- I Used VS Code Development – Lightweight and fast
- Xaamp as a Server – free and easy to handle
- Figma for Wireframe and UI – free and very advance software
- Draw.IO for Diagrams – free and advance support
- Google Docs , Forms for Analysing , Requirment Engineering

- As a hardware used my personal laptop (I5 11th Gen, 24GB RAM)

Performance Optimization Techniques :

- Use Third Party Services and APIs
- Optimized Database Queries and Refactored Code
- Three Tier Architecture
- Lightweight and Advance Tech Stack
- Simple GUI

For Security

- Improved Code level security
- Advance Encryption Techniques
- Validation of SQL Queries, Input Fields

Special Codes

AI Intergration with feature

```

        questions = result.choices[0].message.content.split('\n').filter(q
=> q.trim() !== '');
        displayQuestion();
    }

    function displayQuestion() {
        const questionContainer = document.getElementById('question');
        questionContainer.innerHTML = '';

        if (currentQuestionIndex < questions.length) {
            const questionDiv = document.createElement('div');
            questionDiv.classList.add('question');

            const label = document.createElement('label');
            label.textContent = questions[currentQuestionIndex];
            label.setAttribute('for', `q${currentQuestionIndex + 1}`);

            const input = document.createElement('input');
            input.type = 'text';
            input.id = `q${currentQuestionIndex + 1}`;
            input.name = `q${currentQuestionIndex + 1}`;

            questionDiv.appendChild(label);
            questionDiv.appendChild(input);

            questionContainer.appendChild(questionDiv);
        } else {
            finalizeAnswers();
        }
    }

```

```

    }
  }

  function submitAnswer() {
    const input = document.querySelector('#question input');
    if (input && input.value.trim() !== '') {
      answers[`q${currentQuestionIndex + 1}`] = input.value;
      currentQuestionIndex++;
      displayQuestion();
    } else {
      alert('Please provide an answer before proceeding.');
```

```

    }
  }

  async function finalizeAnswers() {
    const response = await
fetch('https://api.openai.com/v1/chat/completions', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',

    },
    body: JSON.stringify({
      model: 'gpt-4',
      messages: [
        {
          role: 'system',
          content: 'You are a helpful assistant specializing
in mental health analysis and suggestions.'
        },
        {
          role: 'user',
          content: `Analyze the following answers to mental
health questions and provide suggestions in the form of a bulleted list with
current status of mental health : ${JSON.stringify(answers)}`
        }
      ]
    })
  });

  const result = await response.json();
  const suggestions = result.choices[0].message.content;

  const outputDiv = document.getElementById('output');
  outputDiv.style.display = 'block';
  outputDiv.querySelector('ul').innerHTML =
suggestions.split('\n').map(item => `<li>${item}</li>`).join('');
}

```

```

function continueToDashboard() {
    window.location.href = 'user_dashboard.php';
}

// Generate questions on page load
generateQuestions().then(() => {
    document.getElementById('nextButton').disabled = false; // Enable
the "Next" button after questions are loaded
});
</script>

```

Video Call Intergrations

```

<script src="https://meet.jit.si/external_api.js"></script>

<script>
function endChat() {
    if (confirm('Are you sure you want to end the chat?')) {
        window.location.href = 'end_chat.php';
    }
}

document.getElementById("start-video-call").onclick = function() {
    var roomName = "mentalhealthcare" + <?=json_encode($conversation_id) ?>;
    var domain = "meet.jit.si";
    var options = {
        roomName: roomName,
        width: 800,
        height: 1000,
        parentNode: document.querySelector('.container'),
        configOverwrite: {
            startWithAudioMuted: true,
            startWithVideoMuted: true
        },
        interfaceConfigOverwrite: {
            TOOLBAR_BUTTONS: ['microphone', 'camera', 'hangup',
'chat', 'sharedvideo', 'settings']
        }
    };
};

```

```

        var api = new JitsiMeetExternalAPI(domain, options);
        api.executeCommand('displayName', 'User/Doctor');
    };
</script>
</body>
</html>

```

Generate random user names

```

function generateMeaningfulUsername() {
    $adjectives = ['Brave', 'Happy', 'Clever', 'Wise', 'Strong', 'Swift',
    'Bold', 'Gentle', 'Healthy', 'Calm'];
    $nouns = ['Tiger', 'Lion', 'Eagle', 'Phoenix', 'Bear', 'Shark', 'Wolf',
    'Hawk', 'Dragon', 'Leopard'];

    $random_adjective = $adjectives[array_rand($adjectives)];
    $random_noun = $nouns[array_rand($nouns)];

    return $random_adjective . $random_noun;
}

```

Saved Battery Precentage , Ip fo User

```

function getUserIP() {
    if (!empty($_SERVER['HTTP_CLIENT_IP'])) {
        return $_SERVER['HTTP_CLIENT_IP'];
    } elseif (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
        return $_SERVER['HTTP_X_FORWARDED_FOR'];
    } else {
        return $_SERVER['REMOTE_ADDR'];
    }
}

if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $username = generateMeaningfulUsername();
    $ip_address = getUserIP();
    $battery_percentage = isset($_POST['battery']) ? intval($_POST['battery'])
: null;

    $stmt = $pdo->prepare("INSERT INTO users (username, ip_address,
battery_percentage) VALUES (?, ?, ?)");
    $stmt->execute([$username, $ip_address, $battery_percentage]);
}

```

Source Code :

https://drive.google.com/file/d/1_YHDzzmnRVgsAjo6AEfFhFZ8MF5iuoT/view?usp=sharing

Chapter 05

Testing (Quality Assurance)

- Here I Went through Manual QA Process
 - I Validate Each Feature Working Properly
- I Used ,
- Sanity Testing : Validate and Verify Each Function
 - Regression Testing : Verify All System Functions after bug fixes
 - Smoke Testing : Verify Critical Scenario
 - Intergration and System Testing : Verify Feature Combination and Full System .
 - Database Testing : Stress and Volume Testing to Verify Backend and Database
 - UAT Testing : User Acceptance Testing to UI UX Verification
 - Other than that perform a performance and security testing .

Test Cases

figure 5.1

1. User Functionalities

Test Case ID	Test Scenario	Steps	Expected Result
TC001	Anonymous Login	1. Navigate to main page. 2. Click "Get Started".	User is logged in anonymously with random name.
TC002	Read Articles	1. Navigate to "Articles". 2. Click on an article.	Article opens successfully.
TC003	Share Articles	1. Select an article. 2. Click "Share".	Article is shared successfully.

Test Case ID	Test Scenario	Steps	Expected Result
TC004	Chat with Professionals	1. Navigate to chat. 2. Select a doctor. 3. Send a message.	Message is delivered successfully.
TC005	Video Call	1. Click on "Video Call". 2. Allow camera access. 3. Initiate call.	Video call connects successfully.
TC006	AI Suggestions	1. Navigate to AI assessment. 2. Answer questionnaire. 3. Submit.	AI provides mental health analysis.
TC007	Make a Donation	1. Navigate to "Donate". 2. Enter payment details. 3. Confirm payment.	Donation is processed successfully.

2. Doctor Functionalities

figure 5.2

Test Case ID	Test Scenario	Steps	Expected Result
TC008	Doctor Registration	1. Navigate to doctor registration page. 2. Enter valid details.	Doctor account added by admin.
TC009	Doctor Login	1. Enter valid doctor credentials. 2. Click "Login".	Doctor is logged in successfully.
TC010	Provide Help via Chat	1. Navigate to chat. 2. Select a user. 3. Send a message. 4. Reply messages	User receives the message successfully.
TC011	Provide Help via Video Call	1. Click on "Video Call". 2. Allow camera access. 3. Initiate call.	Video call connects successfully.
TC012	Set Availability Status	1. Navigate to doctor dashboard. 2. Change status to "Available" or "Not Available".	Status is updated successfully.

3. Admin Functionalities

figure 5.3

Test Case ID	Test Scenario	Steps	Expected Result
TC013	Admin Login	1. Navigate to admin login page. 2. Enter valid credentials. 3. Click "Login".	Admin is logged in successfully.
TC014	Add/Edit/Delete Articles	1. Navigate to article management. 2. Add/Edit/Delete an article.	Article is updated successfully.

Test Case ID	Test Scenario	Steps	Expected Result
TC015	Manage Users	1. Navigate to user management. 2. Edit/Delete a user.	User is updated or removed successfully.
TC016	Add & Manage Doctors	1. Navigate to doctor management. 2. Approve/Reject doctor registration.	Doctor registration status is updated.

4. Security & Performance Testing

figure 5.4

Test Case ID	Test Scenario	Steps	Expected Result
TC017	Security Testing	1. Attempt SQL injection , Xss ,payload or unauthorized access,.	System prevents security breach.have proper validation
TC018	Database Testing	1. Check data integrity and retrieval. 2. Stress testing	Data is stored and retrieved correctly with advance encryption. Should handle large amout data.
TC019	Performance Testing	1. Simulate multiple users logging and activity in simultaneously.	System handles concurrent users without crashes.
TC020	UAT Testing	1. Conduct end-user testing.	Users confirm the system meets their needs.

These test cases cover **functional, security, database, and performance aspects** to ensure the system is **reliable, efficient, and secure**.

During process found few bugs and fixed all issues .

In this system all test cases are passed .

Chapter 06

Web Link : <http://mentalhealthcare.rf.gd/>

Deployment and Maintenance

- I Deployed it In infinityfree hosting service

6.1

The screenshot shows a web interface for managing MySQL databases. The top section, titled 'MySQL Connection Details', contains fields for MySQL USERNAME (if0_38266121), MySQL PASSWORD (Y5qVCMI05CX5jre5 with a Show/Hide button), MySQL DATABASE NAME (if0_38266121_XXX (see below)), MySQL HOSTNAME (sql305.infinityfree.com), and MySQL PORT (OPTIONAL) (3306). Below this is a section titled 'List of MySQL Databases' which contains a table with columns DATABASE NAME and ACTIONS. The table lists a single database: if0_38266121_mental_health_care, with actions for phpMyAdmin and Delete. At the bottom, there is a '+ Create Database' button and a pagination indicator showing 'Showing 1 to 1 of 1 results' with a page number '1'.

MySQL Connection Details		
MYSQL USERNAME	MYSQL PASSWORD	MYSQL DATABASE NAME
if0_38266121	Y5qVCMI05CX5jre5 Show/Hide	if0_38266121_XXX (see below)
MYSQL HOSTNAME	MYSQL PORT (OPTIONAL)	
sql305.infinityfree.com	3306	

List of MySQL Databases	
DATABASE NAME	ACTIONS
if0_38266121_mental_health_care	phpMyAdmin Delete

[+ Create Database](#) Showing 1 to 1 of 1 results [1](#)

- Also Configured SSL Certificaations

6.2

The screenshot shows a web interface for managing SSL certificates. At the top, there is a section titled 'SSL Certificates' with tabs for All, Pending, Issued, Expired, Revoked, and Failed. Below this is a table with columns DOMAIN, PROVIDER, STATUS, and EXPIRES AT. The table lists two certificates: one for mentalhealthcare.rf.gd issued by Google Trust with a status of Pending, and another for mentalhealthcare.rf.gd issued by Self Signed with a status of Issued and an expiration date of 2035-02-08. Each row has a 'Manage' button. At the bottom, there is a '+ New SSL Certificate' button and a pagination indicator showing 'Showing 1 to 2 of 2 results' with a page number '1'.

SSL Certificates			
All Pending Issued Expired Revoked Failed			
DOMAIN	PROVIDER	STATUS	EXPIRES AT
mentalhealthcare.rf.gd	Google Trust	Pending	-
mentalhealthcare.rf.gd	Self Signed	Issued	2035-02-08

[+ New SSL Certificate](#) Showing 1 to 2 of 2 results [1](#)

Maintenance.

- Future Plan to Engage with NGO , Hospitals and More Professionals
- Start as a Business
- Make a Global Platform
- Improve AI Features More
- User Support and Continues Improvement According Feedback

References

- Andersson, G., Titov, N., Dear, B.F., Rozental, A. and Carlbring, P., 2019. Internet-delivered psychological treatments: from innovation to implementation. *World Psychiatry*, 18(1), pp.20-28.
- Corrigan, P.W., Druss, B.G. and Perlick, D.A., 2014. The impact of mental illness stigma on seeking and participating in mental health care. *Psychological Science in the Public Interest*, 15(2), pp.37-70.
- Naslund, J.A., Aschbrenner, K.A., Marsch, L.A. and Bartels, S.J., 2017. The future of mental health care: peer-to-peer support and social media. *Epidemiology and Psychiatric Sciences*, 25(2), pp.113-122.

Owner – Tharushi Rashi Paba